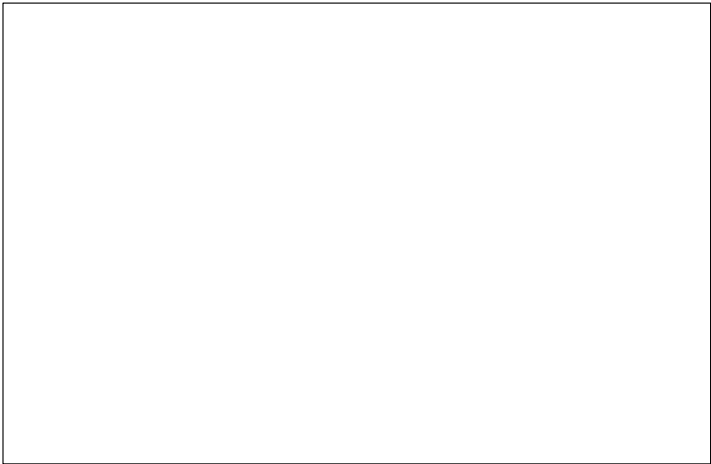


Motor Control Board Fabrication Document

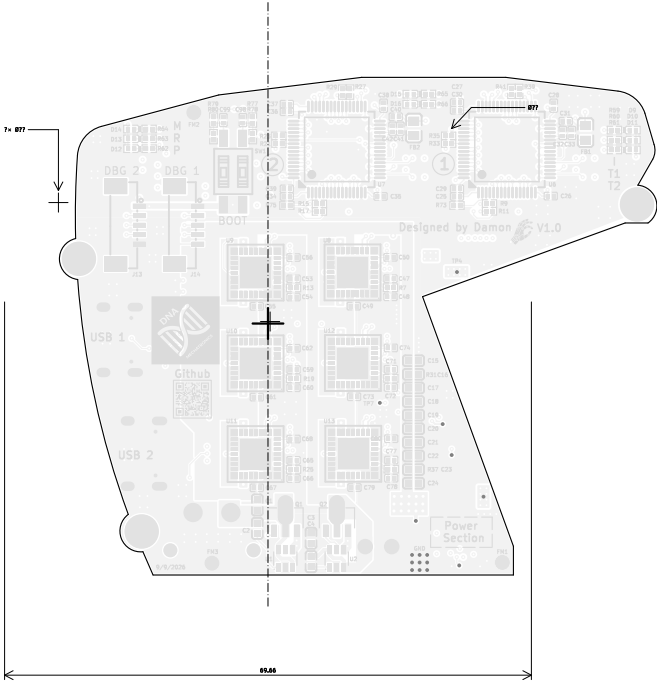
Layer Stack Legend



Impedance Table

| Transmission Line              | Impedance [ohms] | Tolerance [ohms] | Layer | Trace Width [mm] | Gap [mm] | Ref. Layers |
|--------------------------------|------------------|------------------|-------|------------------|----------|-------------|
| Edge-Coupled Coated Microstrip | 100              | ±10 %            | L1    | 0.2032           | 0.28     | L2          |

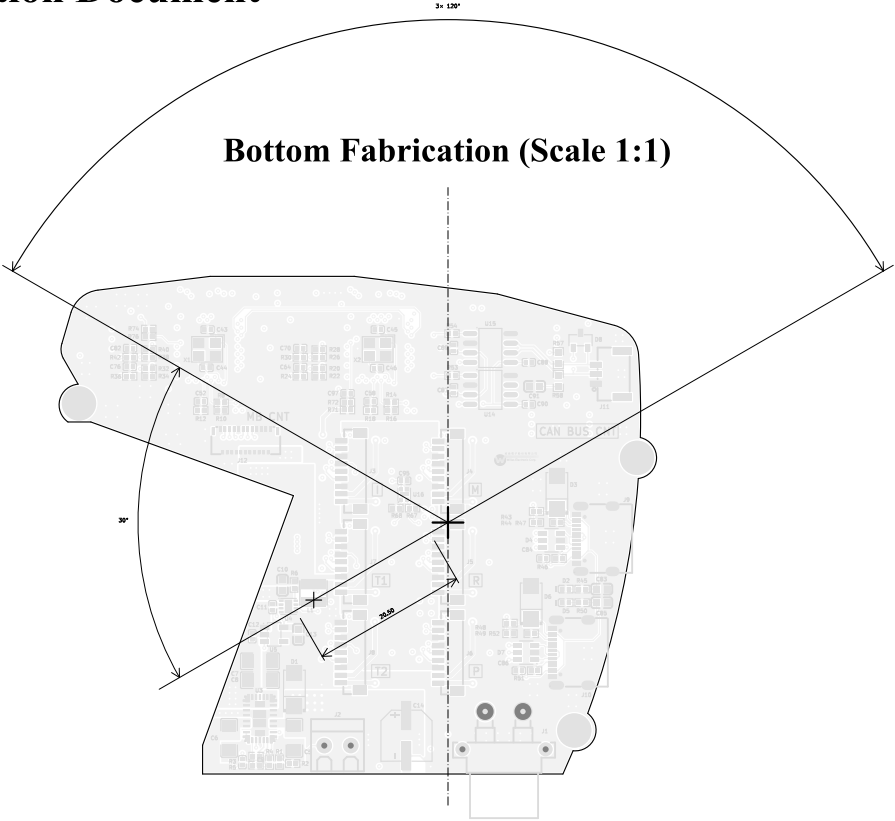
Top Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

|  |                                             |                                           |                    |                              |                             |
|--|---------------------------------------------|-------------------------------------------|--------------------|------------------------------|-----------------------------|
|  | Comments:                                   | Company:<br>Willas                        |                    | Variant:<br>DRAFT            | Git Hash:<br>76c2375        |
|  |                                             | Board Name:<br>Motor Control Board        |                    | Project Name:<br>Master Hand |                             |
|  | Sheet Title:<br>Top Fabrication (Scale 1:1) | File Name:<br>Master_Hand_Kicad.kicad_pcb | Designer:<br>Damon | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                                 | Reviewer:                                 |                    | Size:<br>A4                  | Sheet:<br>1 of 12           |

Motor Control Board Fabrication Document



All dimensions are in millimeters unless otherwise specified.

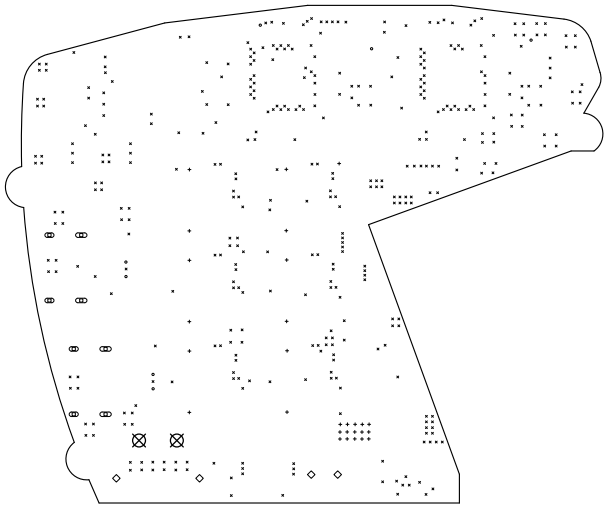
|  |                                                |                                           |                    |                              |                             |
|--|------------------------------------------------|-------------------------------------------|--------------------|------------------------------|-----------------------------|
|  | Comments:                                      | Company:<br>Willas                        |                    | Variant:<br>DRAFT            | Git Hash:<br>76c2375        |
|  |                                                | Board Name:<br>Motor Control Board        |                    | Project Name:<br>Master Hand |                             |
|  | Sheet Title:<br>Bottom Fabrication (Scale 1:1) | File Name:<br>Master_Hand_Kicad.kicad_pcb | Designer:<br>Damon | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                                    |                                           | Reviewer:          | Size:<br>A4                  | Sheet:<br>2 of 12           |

# Motor Control Board Fabrication Document

## Drill Drawing L1 - L6 (Scale 1:1)

Drill Table

| Symbol    | Count | Hole Size          | Plated | Hole Shape | Drill Layer Pair         | Hole Type |
|-----------|-------|--------------------|--------|------------|--------------------------|-----------|
| ×         | 407   | 0,25mm (9,84mils)  | PTH    | Round      | L1 (Sig, PWR) - L6 (GND) | Via       |
| ○         | 7     | 0,30mm (11,81mils) | PTH    | Round      | L1 (Sig, PWR) - L6 (GND) | Via       |
| +         | 28    | 0,40mm (15,75mils) | PTH    | Round      | L1 (Sig, PWR) - L6 (GND) | Via       |
| □         | 8     | 0,60mm (23,62mils) | PTH    | Slot       | L1 (Sig, PWR) - L6 (GND) | Pad       |
| ◇         | 4     | 1,00mm (39,37mils) | PTH    | Round      | L1 (Sig, PWR) - L6 (GND) | Pad       |
| ⊗         | 2     | 1,70mm (66,93mils) | PTH    | Round      | L1 (Sig, PWR) - L6 (GND) | Pad       |
| Total 456 |       |                    |        |            |                          |           |



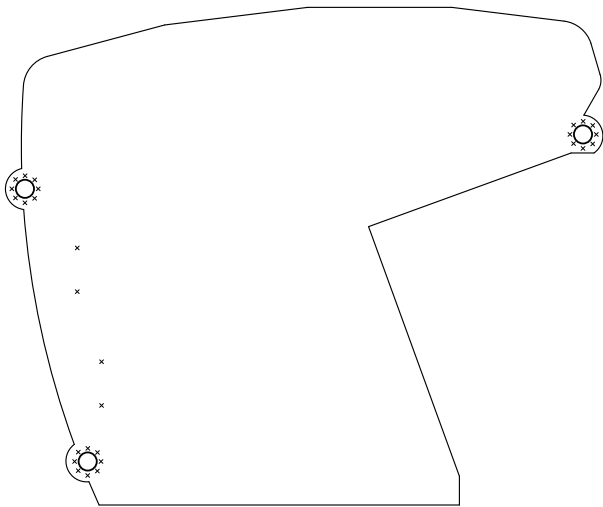
|  |                                         |                                           |                    |                              |                             |
|--|-----------------------------------------|-------------------------------------------|--------------------|------------------------------|-----------------------------|
|  | Comments:                               | Company:<br>Willas                        |                    | Variant:<br>DRAFT            | Git Hash:<br>76c2375        |
|  |                                         | Board Name:<br>Motor Control Board        |                    | Project Name:<br>Master Hand |                             |
|  | Sheet Title:<br>Drill Drawing (L1 - L6) | File Name:<br>Master_Hand_Kicad.kicad_pcb | Designer:<br>Damon | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                             |                                           | Reviewer:          | Size:<br>A4                  | Sheet:<br>3 of 12           |

# Motor Control Board Fabrication Document

## Drill Drawing L1 - L6 (Scale 1:1)

Drill Table

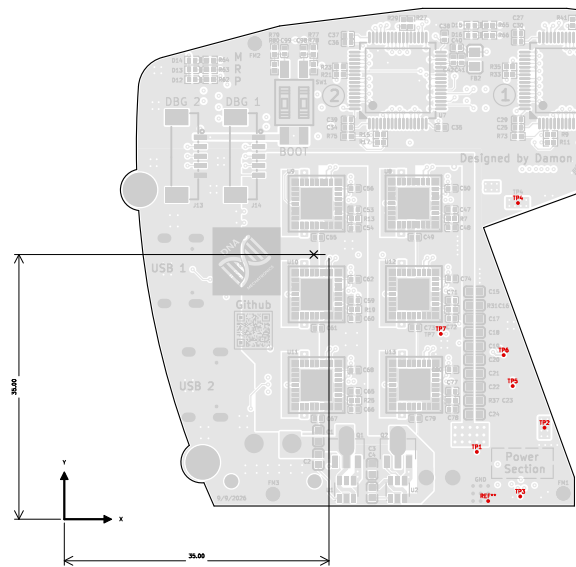
| Symbol | Count    | Hole Size          | Plated | Hole Shape | Drill Layer Pair         | Hole Type  |
|--------|----------|--------------------|--------|------------|--------------------------|------------|
| X      | 28       | 0.50mm (19,69mils) | NPTH   | Round      | L1 (Sig. PWR) - L6 (GND) | Mechanical |
| O      | 3        | 2.40mm (94,49mils) | NPTH   | Round      | L1 (Sig. PWR) - L6 (GND) | Mechanical |
|        | Total 31 |                    |        |            |                          |            |



|  |                                         |                                           |                    |                              |                             |
|--|-----------------------------------------|-------------------------------------------|--------------------|------------------------------|-----------------------------|
|  | Comments:                               | Company:<br>Willas                        |                    | Variant:<br>DRAFT            | Git Hash:<br>76c2375        |
|  |                                         | Board Name:<br>Motor Control Board        |                    | Project Name:<br>Master Hand |                             |
|  | Sheet Title:<br>Drill Drawing (L1 - L6) | File Name:<br>Master_Hand_Kicad.kicad_pcb | Designer:<br>Damon | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                             |                                           | Reviewer:          | Size:<br>A4                  | Sheet:<br>4 of 12           |

# Motor Control Board Fabrication Document

Top Test Points (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

| Ref. | Net | X [mm] | Y [mm] |
|------|-----|--------|--------|
| TP1  |     | -64.00 | 106.00 |
| TP2  |     | -64.00 | 106.00 |
| TP3  |     | -64.00 | 106.00 |
| TP4  |     | -64.00 | 106.00 |
| TP5  |     | -64.00 | 106.00 |
| TP6  |     | -64.00 | 106.00 |
| TP7  |     | -64.00 | 106.00 |
| TP8  |     | -64.00 | 106.00 |
| TP9  |     | -64.00 | 106.00 |
| TP10 |     | -64.00 | 106.00 |
| TP11 |     | -64.00 | 106.00 |
| TP12 |     | -64.00 | 106.00 |
| TP13 |     | -64.00 | 106.00 |
| TP14 |     | -64.00 | 106.00 |
| TP15 |     | -64.00 | 106.00 |
| TP16 |     | -64.00 | 106.00 |
| TP17 |     | -64.00 | 106.00 |
| TP18 |     | -64.00 | 106.00 |
| TP19 |     | -64.00 | 106.00 |
| TP20 |     | -64.00 | 106.00 |
| TP21 |     | -64.00 | 106.00 |
| TP22 |     | -64.00 | 106.00 |
| TP23 |     | -64.00 | 106.00 |
| TP24 |     | -64.00 | 106.00 |
| TP25 |     | -64.00 | 106.00 |
| TP26 |     | -64.00 | 106.00 |
| TP27 |     | -64.00 | 106.00 |
| TP28 |     | -64.00 | 106.00 |
| TP29 |     | -64.00 | 106.00 |
| TP30 |     | -64.00 | 106.00 |
| TP31 |     | -64.00 | 106.00 |
| TP32 |     | -64.00 | 106.00 |

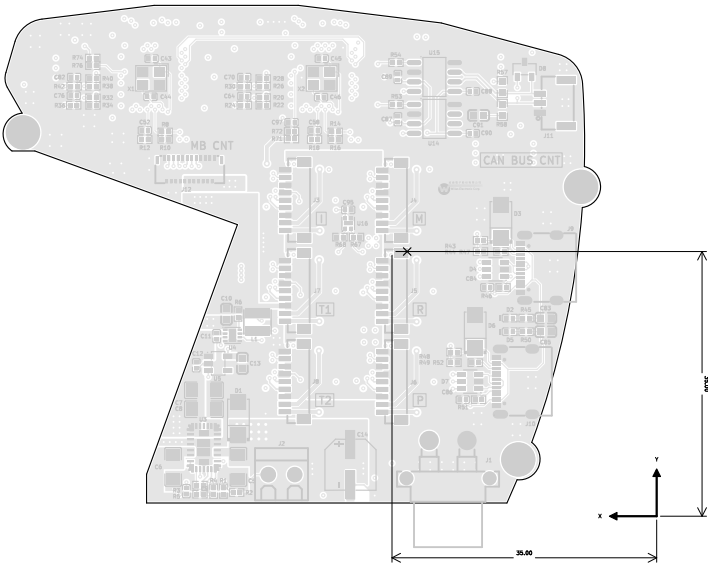
| Ref. | Net | X [mm] | Y [mm] |
|------|-----|--------|--------|
| TP33 |     | -64.00 | 106.00 |
| TP34 |     | -64.00 | 106.00 |
| TP35 |     | -64.00 | 106.00 |
| TP36 |     | -64.00 | 106.00 |
| TP37 |     | -64.00 | 106.00 |
| TP38 |     | -64.00 | 106.00 |
| TP39 |     | -64.00 | 106.00 |
| TP40 |     | -64.00 | 106.00 |
| TP41 |     | -64.00 | 106.00 |
| TP42 |     | -64.00 | 106.00 |
| TP43 |     | -64.00 | 106.00 |
| TP44 |     | -64.00 | 106.00 |
| TP45 |     | -64.00 | 106.00 |
| TP46 |     | -64.00 | 106.00 |
| TP47 |     | -64.00 | 106.00 |
| TP48 |     | -64.00 | 106.00 |
| TP49 |     | -64.00 | 106.00 |
| TP50 |     | -64.00 | 106.00 |

|  |                                             |                                           |                    |                              |                             |
|--|---------------------------------------------|-------------------------------------------|--------------------|------------------------------|-----------------------------|
|  | Comments:                                   | Company:<br>Willas                        |                    | Variant:<br>DRAFT            | Git Hash:<br>76c2375        |
|  |                                             | Board Name:<br>Motor Control Board        |                    | Project Name:<br>Master Hand |                             |
|  | Sheet Title:<br>Top Test Points (Scale 1:1) | File Name:<br>Master_Hand_Kicad.kicad_pcb | Designer:<br>Damon | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                                 |                                           | Reviewer:          | Size:<br>A4                  | Sheet:<br>5 of 12           |

Motor Control Board Fabrication Document

Bottom Test Points (Scale 1:1)

| Ref. | Net | X [mm] | Y [mm] |
|------|-----|--------|--------|
|------|-----|--------|--------|

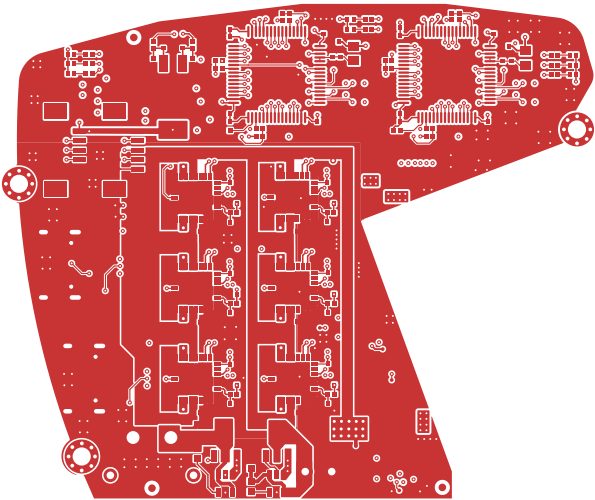


All dimensions are in millimeters unless otherwise specified.

|  |                                                |                                           |                    |                              |                             |
|--|------------------------------------------------|-------------------------------------------|--------------------|------------------------------|-----------------------------|
|  | Comments:                                      | Company:<br>Willas                        |                    | Variant:<br>DRAFT            | Git Hash:<br>76c2375        |
|  |                                                | Board Name:<br>Motor Control Board        |                    | Project Name:<br>Master Hand |                             |
|  | Sheet Title:<br>Bottom Test Points (Scale 1:1) | File Name:<br>Master_Hand_Kicad.kicad_pcb | Designer:<br>Damon | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                                    |                                           | Reviewer:          | Size:<br>A4                  | Sheet:<br>6 of 12           |

# Motor Control Board Fabrication Document

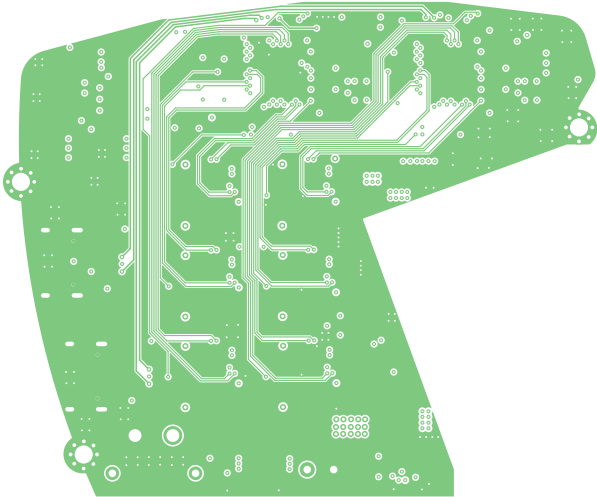
L1 (Sig, PWR) (Scale 1:1)



|  |                                           |                                           |                    |                              |                             |
|--|-------------------------------------------|-------------------------------------------|--------------------|------------------------------|-----------------------------|
|  | Comments:                                 | Company:<br>Willas                        |                    | Variant:<br>DRAFT            | Git Hash:<br>76c2375        |
|  |                                           | Board Name:<br>Motor Control Board        |                    | Project Name:<br>Master Hand |                             |
|  | Sheet Title:<br>L1 (Sig, PWR) (Scale 1:1) | File Name:<br>Master_Hand_Kicad.kicad_pcb | Designer:<br>Damon | Date:<br>2024-04-13          | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                               |                                           | Reviewer:          | Size:<br>A4                  | Sheet:<br>7 of 12           |

# Motor Control Board Fabrication Document

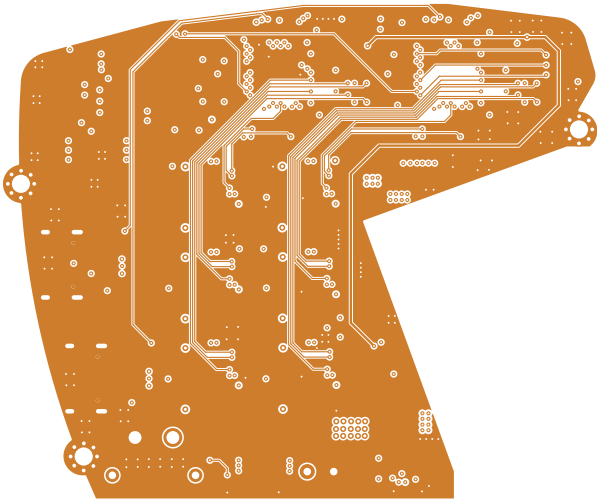
L2 (GND) (Scale 1:1)



|  |                      |                             |           |               |                |
|--|----------------------|-----------------------------|-----------|---------------|----------------|
|  | Comments:            | Company:                    |           | Variant:      | Git Hash:      |
|  |                      | Willas                      |           | DRAFT         | 76c2375        |
|  | Sheet Title:         | Board Name:                 |           | Project Name: |                |
|  |                      | Motor Control Board         |           | Master Hand   |                |
|  | L2 (GND) (Scale 1:1) | File Name:                  | Designer: | Date:         | Revision:      |
|  |                      | Master_Hand_Kicad.kicad_pcb | Damon     | 2024-04-13    | + (Unreleased) |
|  | Sheet Path:          |                             | Reviewer: | Size:         | Sheet:         |
|  |                      |                             |           | A4            | 8 of 12        |

# Motor Control Board Fabrication Document

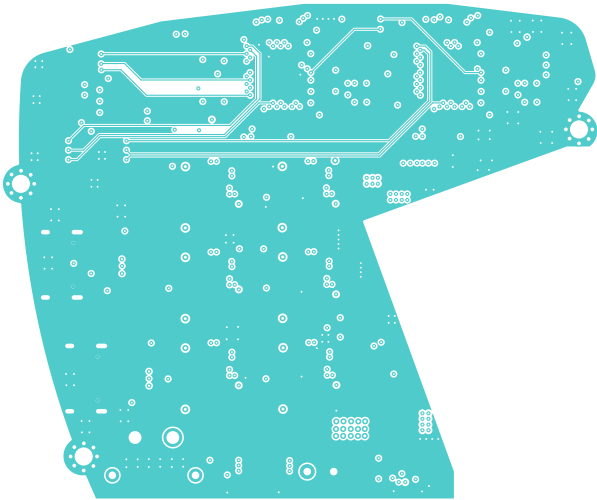
L3 (Sig, PWR) (Scale 1:1)



|  |                           |                             |           |               |                |
|--|---------------------------|-----------------------------|-----------|---------------|----------------|
|  | Comments:                 | Company:                    |           | Variant:      | Git Hash:      |
|  |                           | Willas                      |           | DRAFT         | 76c2375        |
|  | Sheet Title:              | Board Name:                 |           | Project Name: |                |
|  |                           | Motor Control Board         |           | Master Hand   |                |
|  | L3 (Sig, PWR) (Scale 1:1) | File Name:                  | Designer: | Date:         | Revision:      |
|  |                           | Master_Hand_Kicad.kicad_pcb | Damon     | 2024-04-13    | + (Unreleased) |
|  | Sheet Path:               |                             | Reviewer: | Size:         | Sheet:         |
|  |                           |                             |           | A4            | 9 of 12        |

# Motor Control Board Fabrication Document

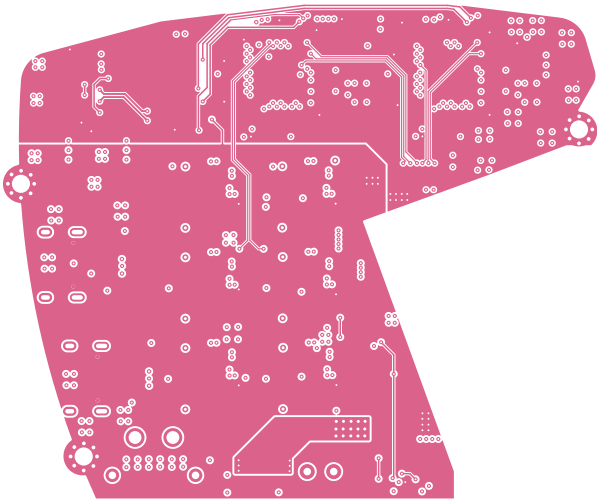
L4 (GND) (Scale 1:1)



|  |                      |                             |           |               |                |
|--|----------------------|-----------------------------|-----------|---------------|----------------|
|  | Comments:            | Company:                    |           | Variant:      | Git Hash:      |
|  |                      | Willas                      |           | DRAFT         | 76c2375        |
|  | Sheet Title:         | Board Name:                 |           | Project Name: |                |
|  |                      | Motor Control Board         |           | Master Hand   |                |
|  | L4 (GND) (Scale 1:1) | File Name:                  | Designer: | Date:         | Revision:      |
|  |                      | Master_Hand_Kicad.kicad_pcb | Damon     | 2024-04-13    | + (Unreleased) |
|  | Sheet Path:          |                             | Reviewer: | Size:         | Sheet:         |
|  |                      |                             |           | A4            | 10 of 12       |

# Motor Control Board Fabrication Document

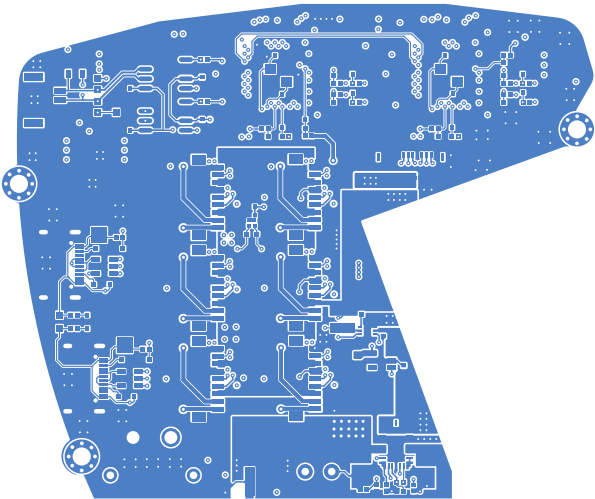
L5 (Sig, PWR) (Scale 1:1)



|  |                           |                     |           |               |           |
|--|---------------------------|---------------------|-----------|---------------|-----------|
|  | Comments:                 | Company:            |           | Variant:      | Git Hash: |
|  |                           | Willas              |           | DRAFT         | 76c2375   |
|  | Sheet Title:              | Board Name:         |           | Project Name: |           |
|  |                           | Motor Control Board |           | Master Hand   |           |
|  | L5 (Sig, PWR) (Scale 1:1) | File Name:          | Designer: | Date:         | Revision: |
|  | Sheet Path:               | Reviewer:           | Size:     | Sheet:        |           |
|  |                           |                     | A4        | 11 of 12      |           |

# Motor Control Board Fabrication Document

L6 (GND) (Scale 1:1)



|  |                      |                     |           |               |           |
|--|----------------------|---------------------|-----------|---------------|-----------|
|  | Comments:            | Company:            |           | Variant:      | Git Hash: |
|  |                      | Willas              |           | DRAFT         | 76c2375   |
|  | Sheet Title:         | Board Name:         |           | Project Name: |           |
|  |                      | Motor Control Board |           | Master Hand   |           |
|  | L6 (GND) (Scale 1:1) | File Name:          | Designer: | Date:         | Revision: |
|  | Sheet Path:          | Reviewer:           | Size:     | Sheet:        |           |
|  |                      |                     | A4        | 12 of 12      |           |